

2023-2024 数理方程 B 试题解答

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1. 设 $u = u(x, y)$, 求解:

$$\begin{cases} u_{xy} + u_y = 0 & (-\infty < x, y < +\infty) \\ u(0, y) = y, & u(x, 0) = x \end{cases}$$

解:

设 $v = u_y$, 则原方程化为:

$$v_x + v = 0$$

解得:

$$v(x, y) = C(y)e^{-x}$$

积分得:

$$u(x, y) = F(y)e^{-x} + D(x) \quad F(y) = \int C(y)dy$$

由初始条件 $u(x, 0) = x$ 得:

$$F(0)e^{-x} + D(x) = x \Rightarrow D(x) = x - F(0)e^{-x}$$

由 $u(0, y) = y$ 得:

$$F(y) + D(0) = y \Rightarrow F(y) = y - D(0) = y - 0 + F(0) = y + F(0)$$

因此解为:

$$u(x, y) = (y + F(0))e^{-x} + x - F(0)e^{-x} = ye^{-x} + x$$

2. 设 $u = u(t, x)$,

$$\begin{cases} u_{tt} = 5u_{xx} + f(t, x) & (t > 0, -\infty < x < +\infty) \\ u|_{t=0} = \sin x, & u_t|_{t=0} = \cos x \end{cases}$$

(1) 如果 $f(t, x) = 0$, 求 $u(t, x)$

(2) 如果 $f(t, x) = \sin(x + t)$, 求 $u(t, x)$.

解:

(1) 当 $f(t, x) = 0$ 时, 方程:

$$\begin{cases} u_{tt} = 5u_{xx} & (t > 0, -\infty < x < +\infty) \\ u|_{t=0} = \sin x, & u_t|_{t=0} = \cos x \end{cases}$$

通解形式:

$$u(t, x) = F(x + \sqrt{5}t) + G(x - \sqrt{5}t)$$

由初始条件:

$$F(x) + G(x) = \sin x$$

$$\sqrt{5}F'(x) - \sqrt{5}G'(x) = \cos x \Rightarrow F(x) - G(x) = \frac{\sin x}{\sqrt{5}} + C$$

解得:

$$F(x) = \left(\frac{1}{2} + \frac{1}{2\sqrt{5}}\right) \sin x + C/2, \quad G(x) = \left(\frac{1}{2} - \frac{1}{2\sqrt{5}}\right) \sin x - C/2$$

因此解为:

$$u(t, x) = \sin x \cos(\sqrt{5}t) + \frac{1}{\sqrt{5}} \cos x \sin(\sqrt{5}t)$$

(2) 当 $f(t, x) = \sin(x + t)$ 时, 因为 (1) 计算了齐次方程通解, 因此接下来只需计算如下非齐次方程特解即可:

$$\begin{cases} v_{tt} = 5v_{xx} + \sin(x + t) & (t > 0, -\infty < x < +\infty) \\ v|_{t=0} = 0, & v_t|_{t=0} = 0 \end{cases}$$

此时可以利用冲量原理, 先求解:

$$\begin{cases} w_{tt} = 5w_{xx} & (t > \tau, -\infty < x < +\infty) \\ w|_{t=\tau} = 0, & w_t|_{t=\tau} = \sin(\tau + x) \end{cases}$$

直接利用达朗贝尔公式可得:

$$w(t, x; \tau) = \frac{1}{2\sqrt{5}} \int_{x-\sqrt{5}(t-\tau)}^{x+\sqrt{5}(t-\tau)} \sin(\tau + \xi) d\xi$$

由此特解形式为:

$$v(t, x) = \int_0^t w(t, x; \tau) d\tau = \frac{1}{2\sqrt{5}} \int_0^t \int_{x-\sqrt{5}(t-\tau)}^{x+\sqrt{5}(t-\tau)} \sin(\xi + \tau) d\xi d\tau$$

计算得:

$$v(t, x) = \frac{1}{4} \sin(x + t) - \frac{1}{4\sqrt{5}} [\cos x \sin(\sqrt{5}t) + \sqrt{5} \sin x \cos(\sqrt{5}t)]$$

由此可得非齐次方程通解为:

$$u(t, x) = \frac{3}{4} \sin x \cos(\sqrt{5}t) + \frac{3}{4\sqrt{5}} \cos x \sin(\sqrt{5}t) + \frac{1}{4} \sin(x + t)$$

二

1. 求解固有值问题的固有值和固有函数:

$$\begin{cases} Y''(x) + \lambda Y(x) = 0, & (-1 < x < 1) \\ Y(-1) = Y(1), & Y'(-1) = Y'(1). \end{cases}$$

解:

(1) $\lambda = 0$: $Y(x) = A + Bx$, 由边界条件得 $B = 0$, 特征函数 $Y_0(x) = 1$.

(2) $\lambda > 0$: 通解 $Y(x) = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$ 。

由边界条件:

$$A \cos \sqrt{\lambda} - B \sin \sqrt{\lambda} = A \cos \sqrt{\lambda} + B \sin \sqrt{\lambda}$$

$$A\sqrt{\lambda} \sin \sqrt{\lambda} + B\sqrt{\lambda} \cos \sqrt{\lambda} = -A\sqrt{\lambda} \sin \sqrt{\lambda} + B\sqrt{\lambda} \cos \sqrt{\lambda}$$

解得 $\sin \sqrt{\lambda} = 0$, 即 $\sqrt{\lambda} = n\pi$ ($n \in \mathbb{Z}^+$)。

特征值 $\lambda_n = (n\pi)^2$, 特征函数 $Y_n(x) = \cos(n\pi x)$ 和 $\sin(n\pi x)$ 。

2. 设 $u = u(r, \theta)$ 求解:

$$\begin{cases} \Delta_2 u = 0 & (1 < r < e, 0 < \theta < \pi) \\ u(r, 0) = 0, & u(r, \pi) = 1, \\ u(1, \theta) = u(e, \theta) = 0 \end{cases}$$

解: 关于 θ 的边界条件非齐次, 需要先利用叠加原理将其齐次化, $u = v + \frac{\theta}{\pi}$, 此时 v 满足如下方程:

$$\begin{cases} \Delta_2 v = 0 & (1 < r < e, 0 < \theta < \pi) \\ v(r, 0) = 0, & v(r, \pi) = 0, \\ v(1, \theta) = v(e, \theta) = -\frac{\theta}{\pi} \end{cases}$$

分离变量 $v(r, \theta) = R(r)\Theta(\theta)$, 得:

$$r^2 R'' + rR' - \lambda R = 0$$

$$\Theta'' + \lambda \Theta = 0$$

由 Θ 的边界条件 $\Theta(0) = 0, \Theta(\pi) = 0$, 得:

$$\Theta_n(\theta) = \sin(n\theta), \quad \lambda_n = n^2$$

R 的解为:

$$R_n(r) = A_n r^n + B_n r^{-n}$$

则 v 的通解为:

$$v = \sum_n (A_n r^n + B_n r^{-n}) \sin(n\theta)$$

带入边界条件:

$$\begin{aligned} \sum_n (A_n + B_n) \sin(n\theta) &= -\frac{\theta}{\pi} \Rightarrow A_n + B_n = \frac{2(-1)^n}{n\pi} \\ \sum_n (A_n e^n + B_n e^{-n}) \sin(n\theta) &= -\frac{\theta}{\pi} \Rightarrow A_n e^n + B_n e^{-n} = \frac{2(-1)^n}{n\pi} \end{aligned}$$

解得

$$A_n = \frac{2(-1)^n}{n\pi(e^n + 1)}, B_n = \frac{2(-1)^n}{n\pi(e^{-n} + 1)}$$

由此可得

$$u = v + \frac{\theta}{\pi} = \frac{\theta}{\pi} + \sum_n \left(\frac{2(-1)^n}{n\pi(e^n + 1)} r^n + \frac{2(-1)^n}{n\pi(e^{-n} + 1)} r^{-n} \right) \sin(n\theta)$$

注：也可以先解 r 方向的固有值方程，做变量代换 $r = e^t$ 原方程变为：

$$\begin{cases} \frac{d^2 R}{dt^2} - \lambda R = 0 \\ R(0) = R(1) = 0 \end{cases}$$

易得固有值 $\lambda = -(n\pi)^2$ ，对应角向固有函数为 $\Theta = C_n \sinh(n\pi\theta) + D_n \cosh(n\pi\theta)$

由此可得 u 的通解为：

$$u = \sum_n (C_n \sinh(n\pi\theta) + D_n \cosh(n\pi\theta)) \sin(n\pi \ln r)$$

带入边界条件可得

$$\sum_n D_n \sin(n\pi \ln r) = 0 \Rightarrow D_n = 0$$

$$\sum_n C_n \sinh(n\pi^2) \sin(n\pi \ln r) = 1 \Rightarrow C_n = \frac{2 - 2(1)^n}{\sinh(n\pi^2)}$$

$$\left(\sinh(n\pi^2) C_n = \frac{\int_1^e \sinh(n\pi \ln r) \frac{1}{r} dr}{\int_1^e \sinh^2(n\pi \ln r) \frac{1}{r} dr} \Rightarrow C_n = \frac{2 - 2(1)^n}{\sinh(n\pi^2)} \right)$$

由此可得

$$u = \sum_n \frac{2 - 2(1)^n}{\sinh(n\pi^2)} \sinh(n\pi\theta) \sin(n\pi \ln r)$$

两种展开方式所得级数的收敛值是一样的，下图展示了 $\theta = \frac{\pi}{4}$ 时两个级数收敛值随 r 的变化：

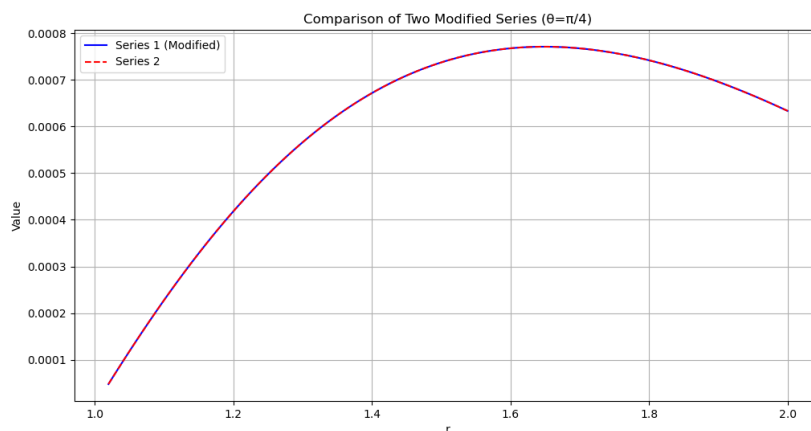


图 1:

三

1. 计算 $\int_0^1 x^3 P_3(x) dx$ 。

解：

$P_3(x) = \frac{1}{2}(5x^3 - 3x)$ ，因此：

$$\int_0^1 x^3 P_3(x) dx = \frac{1}{2} \int_0^1 (5x^6 - 3x^4) dx = \frac{1}{2} \left[\frac{5}{7} - \frac{3}{5} \right] = \frac{2}{35}$$

2. 求解:

$$\begin{cases} \Delta_3 u = 2 & (r < 1, \theta \in (0, \pi)) \\ (u + u_r)|_{r=1} = \cos(2\theta). \end{cases}$$

解:

设 $u = u_0 + u_1$, 其中 $\Delta_3 u_0 = 2$, $\Delta_3 u_1 = 0$ 。

取 $u_0 = \frac{r^2}{3}$, 则边界条件化为:

$$\frac{1}{3} + \frac{2}{3} + u_1(1, \theta) + \partial_r u_1(1, \theta) = \cos(2\theta)$$

即:

$$u_1(1, \theta) + \partial_r u_1(1, \theta) = \cos(2\theta) - 1 = \frac{1}{3}P_2(\cos \theta) - \frac{1}{3}P_0(\cos \theta)$$

u_1 的通解为:

$$u_1(r, \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos \theta)$$

由边界条件得:

$$A_0 = -\frac{1}{3}, A_2 + 2A_2 = \frac{1}{3} \Rightarrow A_2 = \frac{1}{9}$$

其他 A_l 皆为 0 因此解为:

$$u(r, \theta) = \frac{r^2}{3} - \frac{1}{3} + \frac{r^2}{9}P_2(\cos \theta)$$

四

设 $u = u(t, x, y)$, $r = \sqrt{x^2 + y^2}$, 求解以下偏微分方程组:

$$\begin{cases} u_t = \Delta_2 u + \delta(t - r), & t > 0, 0 < r < 1, \\ u|_{r=1} = 0, \\ u(0, x, y) = 0, \end{cases}$$

解答

由于问题具有球对称性, 将方程转换为极坐标 (r, θ) 。忽略 θ 依赖项, 拉普拉斯算子简化为:

$$\Delta_2 u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right).$$

原方程变为:

$$u_t = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \delta(t - r).$$

先求解齐次方程 $u_t = \Delta_2 u$ 的本征函数。设分离变量解 $u(t, r) = T(t)R(r)$, 代入得:

$$T'R = \frac{1}{r}(rR')'T \Rightarrow \frac{T'}{T} = \frac{(rR')'}{rR} = -\lambda.$$

空间部分满足:

$$(rR')' + \lambda rR = 0, \quad R(1) = 0.$$

这是贝塞尔方程, 解为 $R_n(r) = J_0(\alpha_n r)$, 其中 α_n 是 J_0 的第 n 个零点 (即 $J_0(\alpha_n) = 0$), 对应本征值 $\lambda_n = \alpha_n^2$ 。

将 $\delta(t-r)$ 按本征函数展开:

$$\delta(t-r) = \sum_{n=1}^{\infty} \delta_n(t) J_0(\alpha_n r),$$

其中系数 $\delta_n(t)$ 为:

$$\delta_n(t) = \frac{2}{J_1^2(\alpha_n)} \int_0^1 \delta(t-r) J_0(\alpha_n r) r dr = \frac{2t J_0(\alpha_n t)}{J_1^2(\alpha_n)} \Theta(1-t),$$

Θ 为阶跃函数。

将解展开为 $u(t, r) = \sum_{n=1}^{\infty} T_n(t) J_0(\alpha_n r)$, 代入方程得:

$$T_n' + \alpha_n^2 T_n = \delta_n(t), \quad T_n(0) = 0.$$

解得:

$$T_n(t) = \int_0^t e^{-\alpha_n^2(t-\tau)} \delta_n(\tau) d\tau = \frac{2}{J_1^2(\alpha_n)} \int_0^{\min(t,1)} \tau J_0(\alpha_n \tau) e^{-\alpha_n^2(t-\tau)} d\tau.$$

综合上述结果, 解为:

$$u(t, r) = \sum_{n=1}^{\infty} \frac{2J_0(\alpha_n r)}{J_1^2(\alpha_n)} \int_0^{\min(t,1)} \tau J_0(\alpha_n \tau) e^{-\alpha_n^2(t-\tau)} d\tau.$$

五

求解:

$$\begin{cases} u_t = 3u_{xx} + u_y & (t > 0, -\infty < x, y < +\infty) \\ u|_{t=0} = e^{-x^2-y^2}. \end{cases}$$

解:

对函数 $u(t, x, y)$ 关于空间变量 x 和 y 进行傅里叶变换:

$$\hat{u}(t, \xi, \eta) = \mathcal{F}[u(t, x, y)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} u(t, x, y) e^{-i(\xi x + \eta y)} dx dy$$

逆变换为:

$$u(t, x, y) = \mathcal{F}^{-1}[\hat{u}(t, \xi, \eta)] = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \hat{u}(t, \xi, \eta) e^{i(\xi x + \eta y)} d\xi d\eta$$

将原方程 $u_t = 3u_{xx} + u_y$ 两边进行傅里叶变换:

$$\mathcal{F}[u_t] = 3\mathcal{F}[u_{xx}] + \mathcal{F}[u_y]$$

利用傅里叶变换的性质:

$$\mathcal{F}[u_t] = \frac{\partial \hat{u}}{\partial t}, \quad \mathcal{F}[u_{xx}] = -\xi^2 \hat{u}, \quad \mathcal{F}[u_y] = i\eta \hat{u}$$

得到变换后的方程:

$$\frac{\partial \hat{u}}{\partial t} = -3\xi^2 \hat{u} + i\eta \hat{u}$$

解一阶常微分方程:

$$\frac{\partial \hat{u}}{\partial t} = (-3\xi^2 + i\eta) \hat{u}$$

通解为:

$$\hat{u}(t, \xi, \eta) = C(\xi, \eta)e^{-3\xi^2 t + i\eta t}$$

初始条件变换:

$$\hat{u}(0, \xi, \eta) = \mathcal{F}[e^{-x^2 - y^2}] = \pi e^{-(\xi^2 + \eta^2)/4}$$

因此:

$$C(\xi, \eta) = \pi e^{-(\xi^2 + \eta^2)/4}$$

完整解为:

$$\hat{u}(t, \xi, \eta) = \pi e^{-3\xi^2 t - (\xi^2 + \eta^2)/4 + i\eta t}$$

计算逆变换:

$$u(t, x, y) = \frac{\pi}{(2\pi)^2} \iint e^{-(3t + \frac{1}{4})\xi^2 - \frac{\eta^2}{4} + i(\eta t + \xi x + \eta y)} d\xi d\eta$$

分离变量后得到两个高斯积分:

$$\begin{aligned} \int e^{-(3t + \frac{1}{4})\xi^2 + i\xi x} d\xi &= \sqrt{\frac{\pi}{3t + \frac{1}{4}}} e^{-\frac{x^2}{4(3t + \frac{1}{4})}} \\ \int e^{-\frac{\eta^2}{4} + i\eta(y+t)} d\eta &= 2\sqrt{\pi} e^{-(y+t)^2} \end{aligned}$$

最终解为:

$$u(t, x, y) = \frac{1}{\sqrt{1+12t}} \exp\left(-\frac{x^2}{1+12t} - (y+t)^2\right)$$

六

1. 区域 $\Omega = \{(x, y) | y < 2\}$, 求泊松方程第一边值问题的格林函数。

解:

使用镜像法。对于点 $(x_0, y_0) \in \Omega$, 格林函数为:

$$G(x, y; x_0, y_0) = -\frac{1}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-y_0)^2} + \frac{1}{2\pi} \ln \sqrt{(x-x_0)^2 + (y-(4-y_0))^2}$$

2. 求解:

$$\begin{cases} u_{xx} + 2u_x + u + 4u_{yy} = 9 & (x, y) \in \Omega \\ u|_{y=2} = \varphi(x). \end{cases}$$

解:

设 $u(x, y) = e^{-x}v(x, y)$, 则原方程化为:

$$v_{xx} + 4v_{yy} = 9e^x, \quad v|_{y=2} = e^x \varphi(x).$$

做变量代换 $\tilde{y} = y/2$:

$$v_{xx} + v_{\tilde{y}\tilde{y}} = 9e^x, \quad v|_{\tilde{y}=1} = e^x \varphi(x).$$

格林函数为:

$$G(x, \tilde{y}; x_0, \tilde{y}_0) = \frac{1}{2\pi} \left[-\ln \sqrt{(x-x_0)^2 + (\tilde{y}-\tilde{y}_0)^2} + \ln \sqrt{(x-x_0)^2 + (\tilde{y}-(2-\tilde{y}_0))^2} \right].$$

$$G(x, y; x_0, y_0) = \frac{1}{2\pi} \left[-\ln \sqrt{(x-x_0)^2 + \frac{1}{4}(y-y_0)^2} + \ln \sqrt{(x-x_0)^2 + \frac{1}{4}(y-(4-y_0))^2} \right].$$

则

$$\begin{aligned}
v(M) &= \int_V f(M_0) G(M; M_0) dM_0 - \oint_{\partial V} \varphi(M_0) \frac{\partial G}{\partial n} dS_0 \\
v(x, y) &= \int_{-\infty}^{\infty} \int_{-\infty}^2 \frac{9e^{x_0}}{4\pi} \ln \frac{(x-x_0)^2 + \frac{1}{4}(y-(4-y_0))^2}{(x-x_0)^2 + \frac{1}{4}(y-y_0)^2} dx_0 dy_0 + \int_{-\infty}^{\infty} e^{x_0} \varphi(x_0) \frac{2-y}{4\pi \left((x-x_0)^2 + \frac{1}{4}(y-2)^2 \right)} dx_0. \\
u = e^{-x} v &= \int_{-\infty}^{\infty} \int_{-\infty}^2 \frac{9e^{x_0-x}}{4\pi} \ln \frac{(x-x_0)^2 + \frac{1}{4}(y-(4-y_0))^2}{(x-x_0)^2 + \frac{1}{4}(y-y_0)^2} dx_0 dy_0 + \int_{-\infty}^{\infty} e^{x_0-x} \varphi(x_0) \frac{2-y}{4\pi \left((x-x_0)^2 + \frac{1}{4}(y-2)^2 \right)} dx_0.
\end{aligned}$$